

Programme	B. Sc. Mathematics Honours			
Course Code	MAT3MN206			
Course Title	APPLIED MATHEMATICS FOR ECONOMIC ANALYSIS			
Type of Course	Minor			
Semester	III			
Academic Level	200 - 299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Higher Secondary Mathematics			
Course Summary	This course covers differential and difference equations and their economic applications. It explores production functions, including the law of variable proportions, isoquants, and optimization of Cobb-Douglas and CES functions. Additionally, it introduces econometrics, focusing on regression analysis and econometric methodology.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply differential and difference equations to model and solve economic problems.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Analyse production functions to understand the relationship between inputs and outputs, including optimization techniques.	An	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO3	Evaluate econometric models to interpret statistical relationships and economic variables.	E	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Books	1. Introduction to Mathematical Economics , Edward.T.Dowling, Schaum's Outline series, 3 rd edition, TMH. 2. Econometrics and Mathematical Economics , SP singh, AP Parashar, HP singh, S.Chand 3. Basic Economics (4/e), Damodar N Gujarati and Sangeeta, TMH Indian Reprint, 2008.			
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I	Differential and Difference Equations - Text (1) (Chapter 16, 17)		12	Min 15
	1	Differential Equation: definition and concepts		
	2	First order linear differential equation, exact differential equations, integrating factors		
	3	Separation of variables, Economic applications		
	4	Difference equations: definitions and concepts		
	5	First order linear difference equations, Economic applications		
	6	The Cobweb Model, the Harrod model		
II	The Production Function - Text (2) (Chapter 14: sec 14.1-14.9)		10	Min 15
	7	Meaning and nature of production function, the Law of Variable Proportions		
	8	Isoquants, Marginal Rate of Technical Substitution (MRTS)		
	9	Producers' equilibrium, expansion of path.		
	10	The elasticity of substitution, ridge lines and Economic region of production		
III	The Production Function(contd.) and Euler's theorem Text (1&2) (Chapter 14: sec 14.10 to 14.13 of text 2, Chap 6: sec 6.9 &6.10 of Text 1)		14	Min 15
	11	Euler's theorem (Statement only), Euler's theorem and homogenous production function		
	12	Cobb Douglas production function, properties, limitations		
	13	CES production function, properties, advantages, limitations		
	14	Returns to scale, Cobb Web theorem		
	15	Optimization of Cobb Douglas, Optimization of CES production Function		
IV	Econometrics - Text (3) (Pages 1 to 59)		12	Min 15
	16	Introduction to Econometrics		
	17	Statistical v/s deterministic relationships, regression v/s correlation		
	18	Types of data, Measurements of Economic variables		
	19	Methodology of Econometrics		
	20	Two variable regression analysis		
	21	Population regression function (PRF), Stochastic specification of PRF		
	22	Sample regression function (SRF)		
V			12	

	Open Ended Module		
	Matrix solution of Simultaneous Differential and Difference equations, Differentiation of Exponential and Logarithmic functions		
References: 1 Mathematical Analysis for Economists, RGD Allen, MacMillan. 2 Fundamentals of Mathematical Economics, A C Chiang & K Wainwright (4/e,) McGraw Hill 3 Introductory Econometrics: A Modern Approach (6/e), Jeffrey M. Wooldridge, Cengage learning 2016			

Note: 1) Proofs of all the results are exempted for external exam. (2) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	3	2	3	2	3	1	2
CO 2	3	1	3	2	3	2	3	1	2
CO 3	2	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

ONLINE COURSES

(These courses are currently available on the government portal SWAYAM. If they are removed in the future, the board will update the course listings accordingly)

I. The course in brackets, including its course code, is equivalent to the online course specified against it.

1. (MAT1CJ101 Differential Calculus + MAT2CJ101 Integral Calculus)

https://onlinecourses.nptel.ac.in/noc24_ma47/preview

Calculus of One Real Variable

By Prof. Joydeep Dutta | IIT Kanpur

2. (MAT3CJ201 MULTIVARIABLE CALCULUS)

https://onlinecourses.nptel.ac.in/noc24_ma52/preview

Calculus of Several Real Variables

By Prof. Joydeep Dutta | IIT Kanpur

3. (MAT4CJ203 REAL ANALYSIS I)

https://onlinecourses.swayam2.ac.in/cec24_ma01/preview

Real Analysis

By Prof. Surajit Borkotokey | Dibrugarh University

4. (MAT5CJ302 ABSTRACT ALGEBRA I)

https://onlinecourses.nptel.ac.in/noc24_ma50/preview

Introduction to Abstract Group Theory

By Prof. Krishna Hanumanthu | Chennai Mathematical Institute

5. (MAT5CJ303 COMPLEX ANALYSIS I + MAT6CJ304 COMPLEX ANALYSIS II)

https://onlinecourses.nptel.ac.in/noc24_ma60/preview

Complex Analysis

By Prof. Pranav Haridas | Kerala School of Mathematics

6. (MAT8EJ401 Advanced Topology)

https://onlinecourses.nptel.ac.in/noc24_ma74/preview

An Introduction to Point-Set-Topology Part-II

By Prof. Anant R. Shastri | IIT Bombay



7. (MAT8EJ402 PARTIAL DIFFERENTIAL EQUATIONS)

https://onlinecourses.nptel.ac.in/noc24_ma73/preview

Partial Differential Equations

By Prof. Sivaji Ganesh | IIT Bombay

8. (MAT8EJ406 OPERATIONS RESEARCH)

https://onlinecourses.swayam2.ac.in/cec24_ma05/preview

Operations Research

By Professor Bibhas C. Giri | Jadavpur University

II. The following courses are intended to offer students additional credits beyond their regular credits.

1. https://onlinecourses.nptel.ac.in/noc24_ma42/preview

Set Theory and Mathematical Logic

By Prof. Amit Kuber | IIT Kanpur

(For first year students)

2. https://onlinecourses.swayam2.ac.in/cec24_ma17/preview

Logic and Sets

By Mr. Mohamed Nishad Maniparambath | Farook

College, Kozhikode

3. https://onlinecourses.nptel.ac.in/noc24_ma89/preview

A Basic Course in Number Theory

By Prof. Shripad Garge | IIT Bombay



Model Question Papers

First Semester



FIRST SEMESTER B.Sc.(CUFYUGP) DEGREE EXAMINATION

OCTOBER 2024

MAT1CJ101 / MAT1MN100: DIFFERENTIAL CALCULUS

(Credits: 4)

Time: Two Hours

Maximum: 70 Marks

Section A

Answer any number of questions

Each question carries 3 marks

Overall Ceiling 24

1. Determine the domain of the composite function $f \circ g$ of the functions $f(x) = \sqrt{x}$ and $g(x) = x + 1$. Evaluate f at the points $g(3)$ and $f(9)$.
2. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x}$.
3. Does the curve $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so, where?
4. The curve $y = ax^2 + bx + c$ passes through the point $(1, 2)$ and is tangent to the line $y = x$ at the origin. Find a, b and c .
5. Find $\frac{dy}{dx}$ if $2y = x^2 + \sin y$.
6. Find the normal to the curve $x^2 - xy + y^2 = 7$ at the point $(-1, 2)$.
7. Find the absolute extrema of $f(x) = x^{\frac{2}{3}}$ on $[-2, 3]$.
8. If $f'(x) = 0$ at each point of an interval I , then show that $f(x) = C$ for all x in I , where C is a constant.
9. Give an example of a function defined on $[0, 1]$ that has neither a local maximum nor a local minimum value at 0 .
10. Show that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

Section B

Answer any number of questions

Each question carries 6 marks

Overall Ceiling 36

11. Give an equation for the shifted graph of $x = 3y^2$ up 2 and right 3 units. Then sketch the original and shifted graphs together.
12. Is any real number exactly 1 less than its cube? Justify your answer.
13. Define the left-hand limit of a function f at a point x_0 . Give one example.